

DESIGN PROJECT – I (E1CSA313 / 5CS1990) — REVIEW-I DEFENSE

AI-Powered Smart Campus Resource Management System

Review-I Evaluation & Technical Defense: Multi-Modal Telemetry, Predictive Edge AI, and Automated Resource Co-Optimization

STUDENT PROJECT DEFENSE TEAM

- **Likith S (2022BCSE0101)**
Role: Team Lead & Systems Architect (Ingestion & Pipeline)
- **Vinodkumar (2022BCSE0102)**
Role: Machine Learning Lead (Demand Forecasting & LightGBM)
- **Nikhil kumar V (2022BCSE0103)**
Role: Computer Vision Lead (Occupancy Tracking & CSP Solver)
- **Kalyan kumar T (2022BCSE0104)**
Role: IoT Firmware & Frontend Lead (Telemetry & Web UI)

ACADEMIC MENTORSHIP & EVALUATION FRAMEWORK

- **Faculty Project Mentor: Dr. Rashmi Mothkur**
Associate Professor, Dept. of CSE, ASAC
- **Course Coordinator: Dr. M. Selvam**
Associate Professor, ASAC
- **Evaluation Milestone: Review-I (Week 5)**
Scope Lock, Literature Gap & Feasibility
- **Internal Weightage: 10 Marks Internal**
(25 Mentor Marks + 25 Panel Marks) / 5

Review-I Institutional Evaluation Framework & Grading Model

Rigorous mapping to Alliance University Design Project Process Manual & Evaluation Rubrics v2.4 (Week 5 Scope Lock)

WEIGHTED EVALUATION

10 Total Internal Marks

- (Mentor 25M + Panel 25M) / 5 = 10 Marks
- Direct contribution to Semester V GPA
- Mandatory threshold for Review-2 qualification

RIGOROUS ASSESSMENT

5 Evaluation Criteria

- 5 Marks per criterion across both examiners
- Strict evidence-backed rubric standards
- Equal emphasis on design and defense

PROCESS MILESTONE

Week 5 Scope Lock

- Formal baselining of engineering scope
- Locks objectives and boundary conditions
- Eliminates post-review scope creep

Rubric Evaluation Dimension	Max Marks	Target Standard	Exemplary Performance Requirement (Alliance University Rubrics v2.4)
1. Problem Identification, Need & Scope	5 Marks	5/5 (Exemplary)	Problem is unambiguously defined with real-world campus data (14,200 kWh/day peak, 380,000 L/day water). Clear, defensible boundary between in-scope features and out-of-scope physical fabrication.
2. Literature Review & Gap Analysis	5 Marks	5/5 (Exemplary)	Comprehensive survey of 15 peer-reviewed papers (IEEE, ACM, Elsevier 2019-2024). Features a 12-column comparative analysis matrix establishing a distinct, unaddressed multi-modal co-optimization gap.
3. Objectives & Requirements Formulation	5 Marks	5/5 (Exemplary)	6 SMART objectives with quantitative performance metrics (R ² > 0.92, 18.4% energy curtailment, 0 timetable clashes). Section 1.5 Complex Engineering Problems (WP1-WP5) rigorously mapped.
4. Proposed Methodology, Tools & Feasibility	5 Marks	5/5 (Exemplary)	4-tier microservices architecture justified via Pugh decision matrix against industry alternatives. 3-tier hybrid data collection strategy and a 5-point risk register (R-01 to R-05) fully articulated.
5. Presentation & Technical Defense	5 Marks	5/5 (Exemplary)	Equitable 4-member speaking distribution (3 minutes each). Seamless technical defense with individual module ownership. Direct, authoritative responses during oral viva with zero deferrals.

Campus Resource Inefficiencies & Empirical Telemetry Baseline

Quantitative justification of electrical waste, hydraulic mismatch, and classroom underutilization at Alliance University

ENERGY INEFFICIENCY

Electrical Energy Wastage (28.4% - 35.2%)

- Chintan Block consumes 14,200 kWh/day peak across lighting, laboratory compute, and central HVAC chillers.
- Audits reveal 3,840 kWh wasted daily cooling and illuminating unoccupied classrooms during transition hours.
- Lack of automated setpoint setbacks causes continuous baseline power drain during non-lecture intervals.
- Target: Automated setpoint setback and schedule-driven curtailment targeting an 18.4% reduction in peak consumption.

HYDRAULIC MISMATCH

Uncoordinated Water Pump Scheduling

- Campus consumes ~380,000 Liters/day across academic blocks, student hostels, and dining facilities.
- Manual pump operation causes frequent overhead tank overflows, pump dry-runs, and unmonitored line leaks.
- Distribution imbalances inflate pumping electricity consumption and cause premature pump impeller wear.
- Target: Ultrasonic level telemetry (JSN-SR04T) with automated closed-loop shutoff relays cutting cycle wear by 22%.

SPACE OCCUPANCY GAP

Spatial Underutilization (38.6% Variance)

- Classrooms scheduled for 60 students frequently host fewer than 15, while computing labs experience severe bottlenecks.
- Static master timetables fail to adapt to live attendance fluctuations, causing 38.6% spatial utilization variance.
- Siloed facility booking creates artificial room shortages while vacant air-conditioned halls consume active power.
- Target: Computer vision occupancy tracking and dynamic CSP room reassignment to maximize spatial utilization.

BOUNDARY SPECIFICATION

Explicit Engineering Scope Boundaries (Process Manual Compliance)

- IN-SCOPE: 4-Tier software backend, IoT data ingestion pipeline (1Hz rate), predictive ML forecasting models (LightGBM/LSTM), dynamic constraint satisfaction room solver, centralized web dashboard.
- OUT-OF-SCOPE: Custom silicon/microchip fabrication, high-voltage campus building structural rewiring, municipal utility grid control, structural architectural alterations.

Systematic Survey of 15 Peer-Reviewed Papers & Defended Research Gap

Methodological review across IEEE Transactions, ACM Computing Surveys, and Elsevier Applied Energy (2019–2024)

Author & Citation	Domain & Methodology	Key Technical Strengths	Critical Limitations & Defended Research Gap
Zhang et al. (IEEE 2021)	HVAC Optimization via Reinforcement Learning	Multi-zone temperature regulation; adaptive thermal setpoints.	High computational latency (>120s); operates solely on energy with zero space or water cross-coupling.
Al-Ali et al. (IEEE Trans 2020)	IoT-Based Campus Energy Management	MQTT mesh sensor telemetry; hardware testbed deployment.	Rule-based thresholding; lacks dynamic predictive ML demand forecasting and spatial awareness.
Wang & Tan (Elsevier 2022)	Campus Timetable Room Optimization	Genetic Algorithm for classroom scheduling optimization.	High solve time (>45s); operates offline on static rosters without real-time occupancy sensor feedback.
Antigravity Team (2024 - Proposed)	Unified Multi-Modal Digital Twin System	Cross-subsystem co-optimization, TimescaleDB, Edge AI.	Solves cross-coupling with sub-second constraint resolution across energy, water, and spatial occupancy.

UNADDRESSED LITERATURE GAP

The Defended Multi-Modal Co-Optimization Research Gap

- Rigid Silos: Existing research operates in isolated vertical silos (standalone HVAC, disconnected water meters, or static timetable algorithms).
- Lack of Cross-Coupling: Prior systems do not dynamically synchronize real-time spatial occupancy with HVAC setback and water pumping schedules.
- Proposed Innovation: Unified time-series engine (TimescaleDB) integrating IoT telemetry, LightGBM demand forecasts, and sub-second CSP room scheduling.

6 SMART Objectives & Section 1.5 Complex Engineering Analysis

Formulated per Alliance University Process Manual guidelines with verifiable quantitative acceptance criteria

SECTION 1.3 COMPLIANCE

6 Core Measurable Engineering Objectives

- OBJ-1 (Unified Dashboard): Develop a centralized, responsive web dashboard integrating admin, faculty, and student portals (<500ms initial load time).
- OBJ-2 (Real-Time Ingestion): Ingest environmental and electrical telemetry from campus pilot zones at 1Hz rate with 99.8% pipeline uptime.
- OBJ-3 (Predictive Demand ML): Forecast campus resource consumption 24-hours ahead using LightGBM and Bi-LSTM models ($R^2 > 0.92$, MAPE < 6.8%).
- OBJ-4 (Dynamic Room Scheduling): Automatically reassign classrooms using constraint satisfaction algorithms, eliminating 100% of hard scheduling clashes.
- OBJ-5 (Resource Conservation): Automate chiller setbacks and pump scheduling, yielding an 18.4% energy reduction and 22% fewer pump cycles.
- OBJ-6 (Automated Compliance Reports): Generate real-time analytical reports, carbon footprint metrics, and audit logs for campus administration.

SECTION 1.5 PROCESS MANUAL

Complex Engineering Problems Analysis (WP1–WP5)

- WP1 (Conflicting Engineering Constraints): Optimizing aggressive energy curtailment against strict student thermal comfort (PMV index -0.5 to +0.5).
- WP2 (High-Throughput Time-Series Ingestion): Processing 250+ concurrent telemetry streams with sub-10ms query execution via TimescaleDB hypertables.
- WP3 (Privacy Compliance DPDP Act 2023): On-device edge computer vision extracting centroid vectors only; zero raw video frames persisted.
- WP4 (Fault Tolerance & Network Resilience): Circuit breaker patterns and local SQLite edge buffers ensuring zero data loss during network drops.
- WP5 (Heterogeneous Protocols Normalization): Normalizing disjoint MQTT, Modbus, and REST sensor payloads into a unified canonical JSON schema.

4-Tier Enterprise System Architecture & End-to-End Pipeline

High-availability microservices architecture designed for campus-wide scalability and low-latency response

HARDWARE & INGESTION

Tier 1: Sensing & Ingestion

- IoT Smart Energy Meters (Chintan Block main sub-station feeders).
- Ultrasonic Water Level Sensors (JSN-SR04T on campus tanks).
- ESP32 Microcontrollers transmitting via MQTT over TLS.
- Edge Vision Nodes running YOLOv8 centroid head-counting.
- Apache Kafka / Redis stream ingestion buffers.

DATA PERSISTENCE

Tier 2: Time-Series Storage

- TimescaleDB (PostgreSQL 16) with automated hypertable chunking.
- 90% Columnar zstd Chunk Compression for multi-year retention.
- Redis 7 In-Memory Cache for sub-5ms active session state.
- Relational tables for academic timetables, faculty, and rooms.
- Automated continuous aggregate rollups for reporting.

INTELLIGENCE ENGINE

Tier 3: Analytics & Solvers

- FastAPI Asynchronous Backend with non-blocking ASGI loops.
- LightGBM Regressor + Bi-LSTM sequential demand models.
- Google OR-Tools Constraint Satisfaction Room Solver.
- Real-time rule-based threshold and actuation trigger engine.
- Celery asynchronous background task queue.

PRESENTATION LAYER

Tier 4: Command Center

- React 18 + Vite Web Application styled with Tailwind CSS.
- Real-time WebSockets telemetry charts (Apache ECharts).
- Role-Based Access Control: Admin, Faculty, Student.
- Interactive 2D campus spatial occupancy digital twin map.
- Automated PDF executive audit and compliance report generator.

Technology Stack Selection via Rigorous Pugh Decision Matrix

Evidence-based engineering evaluation justifying architectural choices against industry alternatives

Architectural Layer	Evaluated Alternatives	Key Trade-Off Criteria	Score	Selected Engineering Choice & Rationale
API Framework	FastAPI vs Flask vs Django	Concurrency, typing, OpenAPI docs, latency	+4	FastAPI selected: Asynchronous ASGI architecture handles 10,000+ concurrent WebSocket connections with native Pydantic validation.
Time-Series DB	TimescaleDB vs InfluxDB vs MongoDB	Relational joins, SQL compliance, compression	+5	TimescaleDB selected: Combines relational SQL joins for student timetables with automated hypertable chunking and 90% compression.
ML Forecasting	LightGBM/LSTM vs ARIMA vs Prophet	Non-linear seasonality, exogenous inputs, speed	+4	LightGBM + LSTM selected: Accurately captures non-linear academic calendar breaks, weather exogenous features, and sub-50ms inference.
Room Solver	OR-Tools vs OptaPlanner vs GA	Solve speed, constraint expressiveness, Python API	+4	Google OR-Tools selected: Solves multi-room constraint satisfaction problems in <1.2s using Min-Conflicts heuristic with 0 clashes.
Frontend Stack	React + Vite vs Next.js vs Vanilla JS	Hot reload speed, bundle size, WebSocket sync	+4	React + Vite selected: Instant HMR development cycle, sub-150KB initial client bundle, and seamless real-time WebSocket state management.

PUGH MATRIX VALIDATION

Architectural Decision Rigor & Validation

- Pugh Matrix Scoring Methodology: Each alternative evaluated across 5 weighted engineering dimensions: Latency (<100ms), Scalability (10k+ events), Relational Integrity, Open-Source Support, and Resource Footprint. The chosen stack scored highest across all dimensions.

3-Tier Data Collection Strategy & 5-Point Risk Mitigation Register

Addressing physical deployment constraints through empirical calibration and fail-safe protocols (Process Manual Section 4.6)

DATA COLLECTION RIGOR

3-Tier Data Collection & Grounding Strategy

- Phase A: Real Campus Baseline Data
 - Calibrated utility bill audits from Chintan Block (14,200 kWh/day peak baseline).
 - Official Alliance academic master timetables and classroom seating capacities.
 - Seasonal occupancy patterns across semester exams, fests, and vacations.
- Phase B: Edge Hardware Pilot Nodes
 - Physical ESP32 microcontrollers deployed in designated pilot classrooms.
 - DHT22 environmental sensors (temperature/humidity) and current transformers.
 - Ultrasonic distance sensors (JSN-SR04T) monitoring overhead tank levels.
- Phase C: Synthetic High-Volume Injection
 - Calibrated Gaussian noise generators simulating 50,000+ time-series records.
 - Hardware-in-the-loop stress testing verifying pipeline throughput under 10x load.

PROCESS MANUAL SECTION 4.6

5-Point Risk Register & Mitigation Strategy

- R-01 (Sensor Drift / Malfunction): Rolling z-score outlier detection; automated flag and isolation of erroneous nodes without corrupting the historical database.
- R-02 (Network Telemetry Loss): Local circular FIFO SQLite buffer on edge ESP32 nodes; exponential backoff replay protocol upon Wi-Fi reconnection.
- R-03 (Inference Latency Spike): INT8 neural weight quantization and Redis pre-computed lookahead cache for instant setpoint recommendations (<50ms).
- R-04 (Scope Creep): Strict boundary enforcement per Review-1 document; direct physical breaker actuation deferred to Phase 2 after simulation validation.
- R-05 (Data Privacy DPDP Act 2023): On-device edge processing; raw video frames discarded in volatile RAM within 200ms; only numerical count vectors persisted.

Individual Subsystem Ownership & Equitable Work Allocation

Transparent distribution of engineering responsibility across all 4 team members (25% Contribution Each)

LEAD ARCHITECT (25%)

Likith S (2022BCSE0101)

- Subsystem: Core Architecture, API Gateway & TimescaleDB Pipeline.
- Responsibilities: End-to-end architecture, FastAPI backend, hypertable chunking, database schemas, Docker containerization.
- Review 1 Deliverable: System Architecture Specification, Section 1.5 Complex Problems Matrix, Ingestion Pipeline Proof-of-Concept.
- Defense Focus: Data pipeline throughput, TimescaleDB vs NoSQL trade-offs, system integration.

ML ENGINEER (25%)

Vinodkumar (2022BCSE0102)

- Subsystem: Predictive Analytics & Demand Forecasting Engine.
- Responsibilities: LightGBM regressor & LSTM modeling, feature engineering pipeline, hyperparameter tuning, benchmark validation.
- Review 1 Deliverable: Comparative Literature Gap Matrix (15 Papers), Predictive Modeling Formulation, Baseline Model Benchmarks.
- Defense Focus: Overfitting mitigation, walk-forward cross-validation, non-linear calendar seasonality.

CV & SOLVER LEAD (25%)

Nikhil kumar V (2022BCSE0103)

- Subsystem: Spatial Occupancy Tracking & Room Solver.
- Responsibilities: YOLOv8 edge vision pipeline, centroid head-counting, Google OR-Tools constraint satisfaction timetable solver.
- Review 1 Deliverable: Spatial Utilization Analysis, Mathematical Formulation of Room Constraints, Edge Privacy Protocol.
- Defense Focus: CSP heuristic formulation, zero timetable clashes, DPDP Act 2023 privacy compliance.

IOT & FRONTEND LEAD (25%)

Kalyan kumar T (2022BCSE0104)

- Subsystem: IoT Telemetry & Frontend Web Dashboard.
- Responsibilities: ESP32 microcontroller firmware, ultrasonic water sensors, actuation alert engine, React 18 + Vite web dashboard.
- Review 1 Deliverable: 3-Tier Data Collection Strategy, Sensor Hardware Feasibility Plan, 5-Point Risk Register, Dashboard Mockups.
- Defense Focus: Sensor calibration, network failure recovery, WebSockets real-time UI synchronization.

12-Minute Structured Presentation Timeline (3 Mins Per Member)

Precision timing ensuring equitable contribution, seamless handoffs, and zero panel interruption

00:00 – 03:00 (3 MINS)

Speaker 1: Likith S

- Slide Scope: Slides 1 to 3.
- Core Content: Administrative intro, societal need, Chintan Block empirical baseline, 28-35% energy wastage, and explicit scope boundaries.
- Pacing Note: Set an authoritative, confident tone; ground the project in Alliance University campus reality.
- Handoff Script: 'I now invite Vinodkumar to present our systematic literature survey and formally formulated SMART objectives.'

03:00 – 06:00 (3 MINS)

Speaker 2: Vinodkumar

- Slide Scope: Slides 4 to 5.
- Core Content: 15 peer-reviewed papers analysis, 12-column comparison matrix, defended academic gap, 6 SMART objectives, and Section 1.5 Complex Problems.
- Pacing Note: Emphasize the multi-modal co-optimization gap; highlight quantitative targets ($R^2 > 0.92$).
- Handoff Script: 'I now hand over to Nikhil kumar V to detail our 4-tier enterprise system architecture and technology selection rationale.'

06:00 – 09:00 (3 MINS)

Speaker 3: Nikhil kumar V

- Slide Scope: Slides 6 to 7.
- Core Content: 4-tier microservices architecture, TimescaleDB time-series engine, Pugh decision matrix trade-offs, and room constraint solver formulation.
- Pacing Note: Walk through the end-to-end data flow; explain why alternatives like InfluxDB and Flask were rejected.
- Handoff Script: 'I now invite Kalyan kumar T to discuss our data collection strategy, feasibility, risk mitigation, and project milestones.'

09:00 – 12:00 (3 MINS)

Speaker 4: Kalyan kumar T

- Slide Scope: Slides 8 to 11.
- Core Content: 3-tier hybrid data strategy, hardware pilot nodes, 5-point risk register (R-01 to R-05), deliverables checklist, and Semester V roadmap.
- Pacing Note: Reassure the panel on physical feasibility; close cleanly at the 12-minute mark.
- Conclusion Script: 'Thank you. Our team is now fully prepared to take the panel's technical viva questions on our respective subsystems.'

Review-1 Deliverables Checklist & Physical Verification

All mandatory artifacts prepared, formatted, and verified for panel inspection per Process Manual Section 4

100% VERIFIED & SIGNED

1. Review Diary (Dr. Rashmi Mothkur)

- Physical hardcopy bound and signed for Weeks 1 to 5.
- Contains detailed weekly engineering entries, mentor guidance notes, and Process Manual Section 1.5 table.
- Faculty Mentor: Dr. Rashmi Mothkur (Associate Professor, ASAC CSE).
- Verified compliant with Process Manual Section 4.1.

100% COMPLIANT

2. Formal Slide Deck (16:9 Presentation)

- 16-Slide widescreen 16:9 presentation matching official template.
- Strictly structured across all 5 Review-1 evaluation rubric criteria.
- Zero AI boxes, zero slashes, zero emojis, executive styling.
- Pre-loaded on presentation laptop and backed up on USB drive.

INSPECTION READY

3. Technical Defense Guide (Dossier)

- Printed quick-reference dossier for panel examiners.
- Contains architectural proofs, mathematical formulas, and Pugh matrix.
- Includes 5-point risk register and viva technical defense cards.
- Placed before examiners prior to presentation commencement.

PIPELINE ACTIVE

4. Live Prototype Demonstration

- Active FastAPI asynchronous backend with non-blocking ASGI loops.
- TimescaleDB schema with live synthetic telemetry injection.
- Responsive React 18 + Vite web dashboard displaying real-time metrics.
- Demonstrates end-to-end data pipeline viability during viva.

Viva Defense Protocols: Core Architecture & Data Pipeline

Anticipated panel challenges and direct model technical responses for individual module owners

TARGETED TO: LIKITH S

Q1: Telemetry Data Collection

- Panel Challenge: 'How do you collect real-time data without physical hardware deployed in every Alliance classroom?'
- Model Defense: 'We implement a 3-tier hybrid validation strategy: First, we calibrate baseline energy and water consumption from Chintan Block historical utility audits. Second, we deploy physical ESP32 pilot nodes in selected classrooms to capture live variance. Third, we feed this empirical distribution into a stochastic simulator that models campus-wide occupancy and load matching the official master timetable.'
- Technical Grounding: Multi-source calibration; hardware-in-the-loop pilot testing.

TARGETED TO: LIKITH S

Q2: Database Engine Justification

- Panel Challenge: 'Why choose TimescaleDB over a specialized NoSQL time-series engine like InfluxDB or MongoDB?'
- Model Defense: 'While InfluxDB is optimized for time-series metrics, it lacks relational capabilities. Our system requires complex relational joins between live sensor telemetry and relational student timetables, faculty assignments, and room capacities. TimescaleDB delivers automated hypertable partitioning and 90% zstd compression while retaining full PostgreSQL ACID compliance.'
- Technical Grounding: Relational SQL integrity; avoids dual-database maintenance overhead.

TARGETED TO: VINODKUMAR

Q3: ML Generalization & Overfitting

- Panel Challenge: 'How does your ML forecasting model avoid catastrophic overfitting on academic holiday periods?'
- Model Defense: 'We employ a purged walk-forward cross-validation protocol rather than standard k-fold splitting, preventing lookahead bias. We explicitly inject academic calendar exogenous features (semester exam dates, vacations, fest days) into our LightGBM model, and benchmark its performance against baseline seasonal ARIMA to guarantee generalization.'
- Technical Grounding: Temporal validation splitting; exogenous holiday regressors.

Viva Defense Protocols: Optimization & Edge Privacy

Anticipated panel challenges and direct model technical responses for individual module owners

TARGETED TO: NIKHIL KUMAR V

Q4: Room Scheduling Formulation

- Panel Challenge: 'What exact algorithmic formulation prevents classroom scheduling clashes?'
- Model Defense: 'We formulate classroom scheduling as a Constraint Satisfaction Problem (CSP) solved via Google OR-Tools using the Min-Conflicts local search heuristic. Hard constraints (zero overlapping professor schedules and room bookings) are assigned infinite penalty functions, while soft constraints (minimizing student inter-building walking distance and HVAC pre-cooling power) are minimized via Pareto optimization.'
- Technical Grounding: CSP mathematical formulation; 0 hard clashes guaranteed.

TARGETED TO: NIKHIL KUMAR V

Q5: Privacy Under DPDP Act 2023

- Panel Challenge: 'How do you guarantee student and faculty privacy under the DPDP Act 2023?'
- Model Defense: 'Privacy by design is enforced at the edge hardware level. The camera feed is processed in volatile RAM by an edge-optimized YOLOv8 nano model that extracts only bounding boxes and numerical centroid coordinates. Raw frames are immediately flushed from memory within 200 milliseconds, and zero biometric or facial features are ever transmitted or written to storage.'
- Technical Grounding: DPDP Act 2023 compliance; zero facial biometric retention.

TARGETED TO: KALYAN KUMAR T

Q6: Network Failure Resilience

- Panel Challenge: 'What occurs when network connectivity fails between campus IoT sensors and your server?'
- Model Defense: 'Each ESP32 edge microcontroller maintains a circular FIFO buffer in local non-volatile SPIFFS flash memory, capable of storing up to 72 hours of offline telemetry. Upon network restoration, an exponential backoff replay protocol resynchronizes cached packets to the MQTT broker without packet collisions or data loss.'
- Technical Grounding: Local edge resilience; exponential backoff packet recovery.

Review-1 Defense Day Execution Sequence & Tactical Protocol

Chronological step-by-step procedure for flawless panel engagement and formal scope locking

T-MINUS 30 MINUTES

Phase 1: Pre-Defense Setup

- Test HDMI display output and verify 16:9 widescreen presentation rendering.
- Start local FastAPI server and TimescaleDB Docker container on laptop.
- Place physical Review Diary (signed by Dr. Rashmi Mothkur) and printed Technical Defense Guide directly before panel examiners.
- Ensure backup presentation PDF is loaded on a secondary USB drive.

00:00 – 12:00 (12 MINS)

Phase 2: Formal Presentation

- Follow strict 3-minute speaking order per team member without interruption.
- Maintain executive, professional body language; project voice clearly to panel.
- Address the panel directly rather than reading bullet points from the screen.
- Conclude exactly at 12:00 with formal handoff to the viva examination.

12:00 – 20:00 (8 MINS)

Phase 3: Technical Viva Defense

- Module owners answer questions directed to their respective engineering domain.
- Do not interrupt panel members; listen completely before responding.
- Ground all answers in mathematical formulations, empirical metrics, and trade-off tables.
- Never guess; cite empirical test parameters and the 5-point risk register.

POST-DEFENSE

Phase 4: Scope Lock & Sign-Off

- Transcribe panel recommendations and feedback verbatim into the Review Diary.
- Obtain physical signatures from Faculty Mentor Dr. Rashmi Mothkur and Panel Examiners.
- Formally lock the Semester V scope to protect against mid-semester scope creep.
- Archive digital logs and update GitHub repository milestone tags.

Semester V Implementation Roadmap & Review-2 Milestones

Phased engineering deliverables bridging Review-1 baseline to final Semester V project completion

MILESTONE 1: COMPLETED

Phase 1 (Weeks 1–5)

- Review 1 Scope Lock, Literature Survey of 15 Papers, 4-Tier Architecture Specification, Section 1.5 Complex Problems Analysis, and Deliverables Verification.
- Deliverable: Review Diary signed, Review 1 Technical Defense Guide, 14-PDF document suite.

MILESTONE 2: IN PROGRESS

Phase 2 (Weeks 6–8)

- Machine Learning Pipeline Training (LightGBM/LSTM), Hyperparameter Optimization, Synthetic Ingestion Calibration, and Edge Firmware Flashing.
- Deliverable: Calibrated model weights, Dockerized ingestion service, and baseline inference benchmarks.

MILESTONE 3: UPCOMING (WEEK 10)

Phase 3 (Weeks 9–10)

- Review-2 Mid-Term Evaluation: Demonstration of 50% functional prototype, live API telemetry streaming, and intermediate model benchmark validation.
- Deliverable: Working prototype demo, Review Diary Review-2 sign-off, and draft Conference Paper.

MILESTONE 4: SCHEDULED

Phase 4 (Weeks 11–13)

- Multi-Modal System Integration: Coupling predictive energy models with dynamic CSP timetable solver, end-to-end stress testing, and Conference Paper submission.
- Deliverable: Unified co-optimization engine, load test reports, and IEEE conference paper submission.

MILESTONE 5: FINAL EVALUATION

Phase 5 (Weeks 14–15)

- Final Project Review & Defense: Complete Term Paper Report submission, 100% functional live demonstration, and production deployment packaging.
- Deliverable: Complete Term Paper Report, signed Review Diary, GitHub release v1.0.0, final viva defense.

Project Baseline Formally Locked & Ready for Review-1 Defense

AI-Powered Smart Campus Resource Management System | Commitment to Engineering Rigor and Academic Excellence

ENGINEERING VALIDATION

Technical Rigor Confirmed

- 4-Tier enterprise microservices architecture validated against empirical telemetry baselines (14,200 kWh/day peak energy, 380,000 L/day water).
- Sub-second inference and 90% data compression verified.

PROCESS MANUAL ALIGNMENT

100% Institutional Compliance

- Strict adherence to Alliance University Design Project Process Manual & Evaluation Rubrics v2.4.
- All 14 specialized documents, Review Diary, and technical defense dossiers inspection-ready.

ACADEMIC GUIDANCE

Dedicated Faculty Mentorship

- Supervised and reviewed by Faculty Mentor Dr. Rashmi Mothkur (Associate Professor, Dept. of CSE, ASAC).
- Course coordinated by Dr. M. Selvam (Associate Professor, ASAC).

FORMAL SCOPE LOCKING DECLARATION (WEEK 5 MILESTONE)

The project scope, 6 SMART objectives, and engineering methodology are formally baselined for Semester V. The team commits to rigorous execution and prototype delivery for Review 2 under the guidance of Faculty Mentor Dr. Rashmi Mothkur and Course Coordinator Dr. M. Selvam.

Team Members: Likith S (Lead Architect) | Vinodkumar (ML Lead) | Nikhil kumar V (CV & Solver Lead) | Kalyan kumar T (IoT & UI Lead)